

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 09

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Recap of Previous Lecture



Topic

TCP Congestion Control





Topics to be Covered



Topic

TCP Congestion Control

Topic

MSL

Topic

TCP Connection Close

[MCQ]

[GATE-IT-2005] [2 Mark]



#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is _____.

- ☐ A 2048 bytes
- ☒ B 4096 bytes
- ☐ C 6144 bytes
- ☐ D 8192 bytes

Ans: B

Congestion Window (CWND) = 4 KB

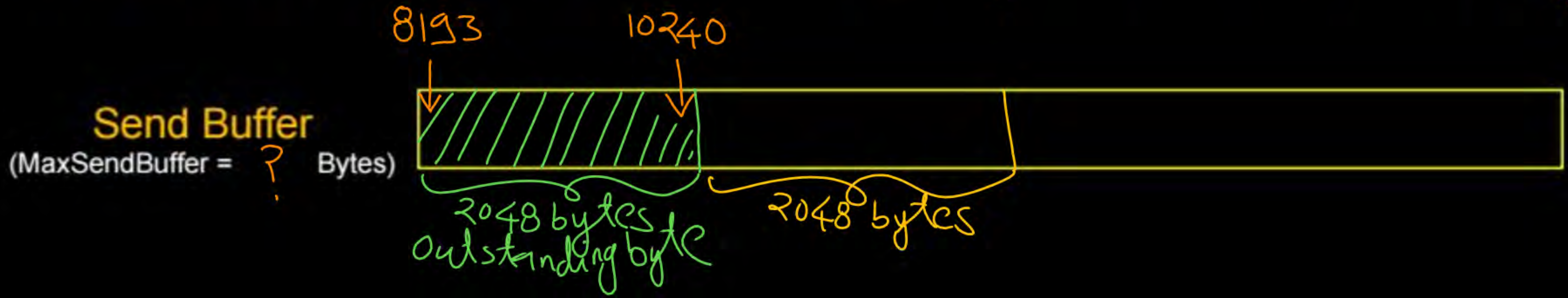
Receiver Advertise Window (RWND) = 6 KB

LastByteSent = 10240

LastByteAcked = 8192

$$[\text{LastByteSent} - \text{LastByteAcked}] \leq \text{minimum}[\text{CWND}, \text{RWND}]$$

Current window size at the sender = $\max^m$ bytes can be outstanding



[NAT]

[GATE-2014][2 Mark]



#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____.

CWND $\rightarrow$ (in bytes)
 $\downarrow$
 $\rightarrow$ (in MSS)

RWND = ? (large)

$$\text{CWND} = 32 \text{ KB} \checkmark$$

$$\text{MSS} = 2 \text{ KB}$$

$$\text{CWND} = 32 \text{ KB} / 2 \text{ KB}$$

$$\text{CWND} = 16 \text{ MSS}$$

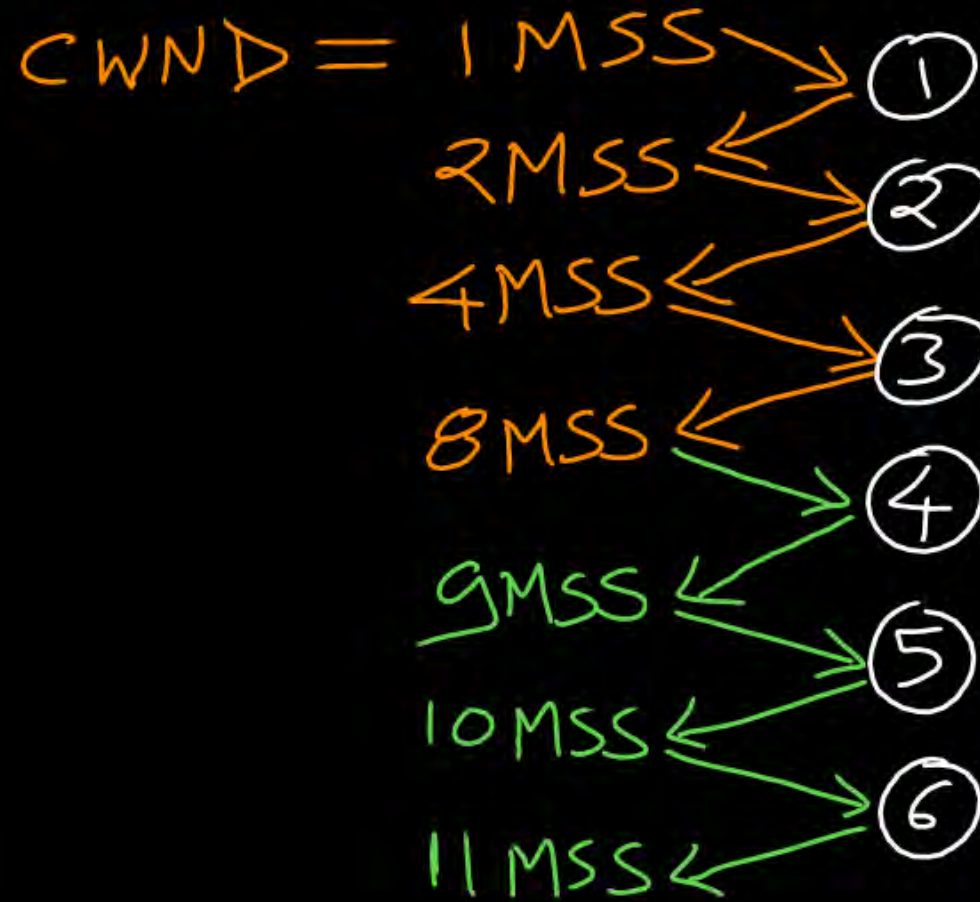
Time-out $\leftarrow$

$$\text{SStresh} = \text{CWND} / 2$$

$$= 16 \text{ MSS} / 2$$

$$\text{SStresh} = 8 \text{ MSS}$$

$$\text{RTT} = 100 \text{ ms}$$



$$\begin{aligned} \text{Ans} &= 11 * \text{RTT} \\ &= 11 * 100 \text{ ms} \\ &= 1100 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{IIT-KGP Ans} \\ &= [1100 - 1300] \end{aligned}$$



[NAT]

[GATE-2020][2 Mark]



#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t=0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t+60$ ms after all acknowledgments are processed is _____ . 

$$\text{RWND} = 50 \text{ KB}$$

$$\text{MSS} = 2 \text{ KB}$$

$$\text{RWND} = 50 \text{ KB} / 2 \text{ KB}$$

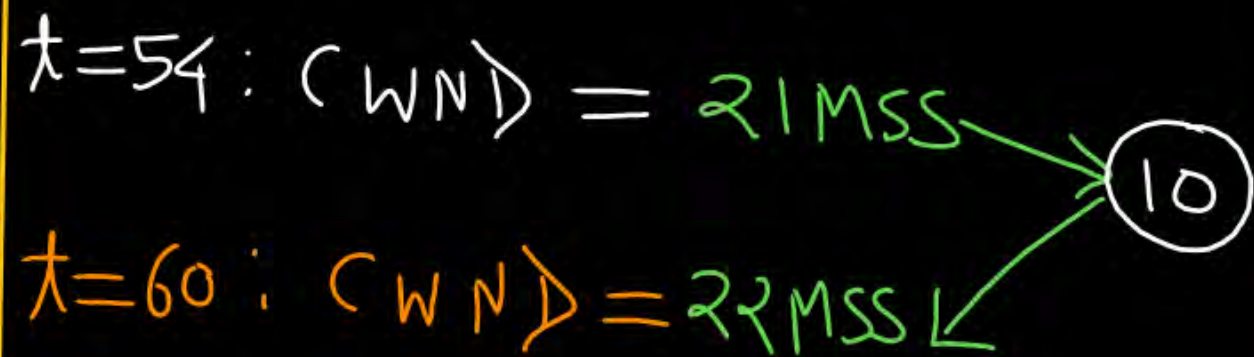
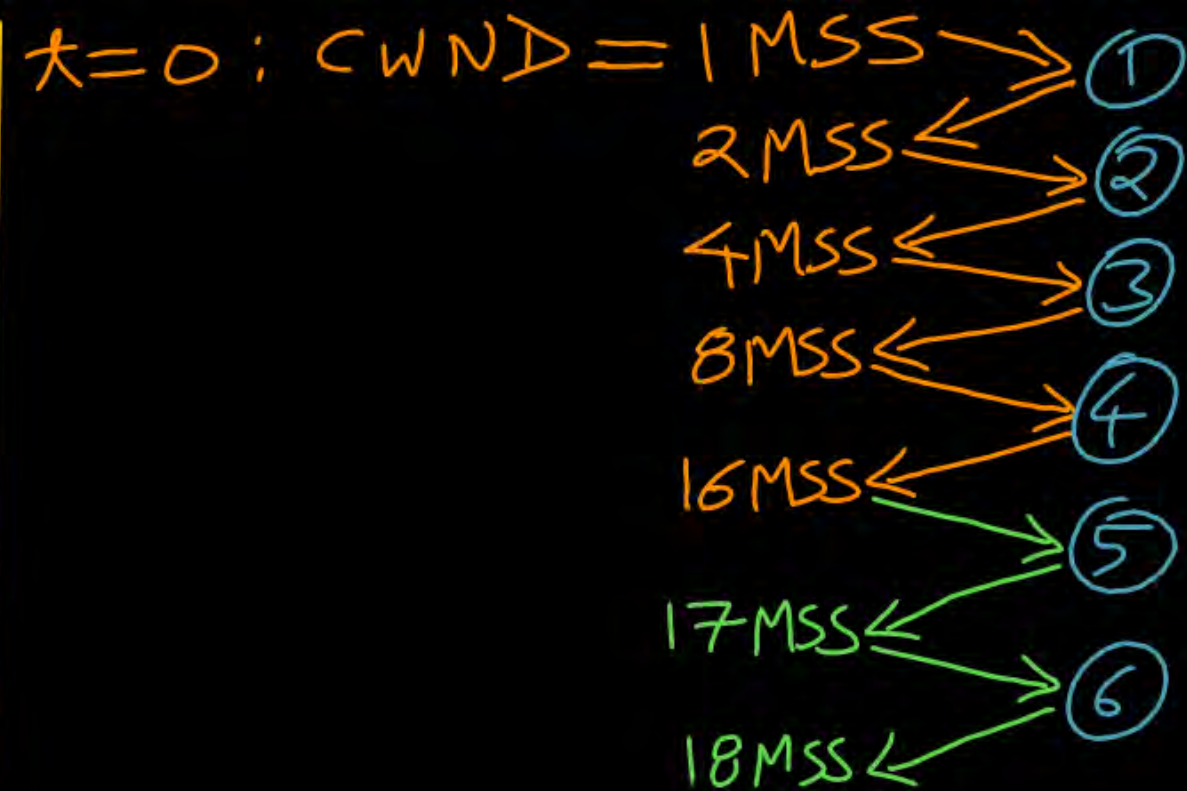
$$\text{RWND} = 25 \text{ MSS} \checkmark$$

$$\text{SStresh} = 32 \text{ KB}$$

$$\text{SStresh} = 32 \text{ KB} / 2 \text{ KB}$$

$$\text{SStresh} = 16 \text{ MSS}$$

$$\text{RTT} = 6 \text{ ms}$$



$$\begin{aligned} \text{Ans} &= 22 \text{ MSS} \\ &= 22 \times 2 \text{ KB} \\ &= 44 \text{ KB} \end{aligned}$$



$$\boxed{\text{Ans} = 44}$$

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

☐ A 7 MSS

☐ B 8 MSS

☒ C 12 MSS

☒ D 14 MSS

TCP Congestion Control Algo.

$\Rightarrow$ AIMD

When Time-out occur

$$ssthresh = \frac{CWND}{2};$$

// Multiplicative Decrease

In Congestion Avoidance Phase

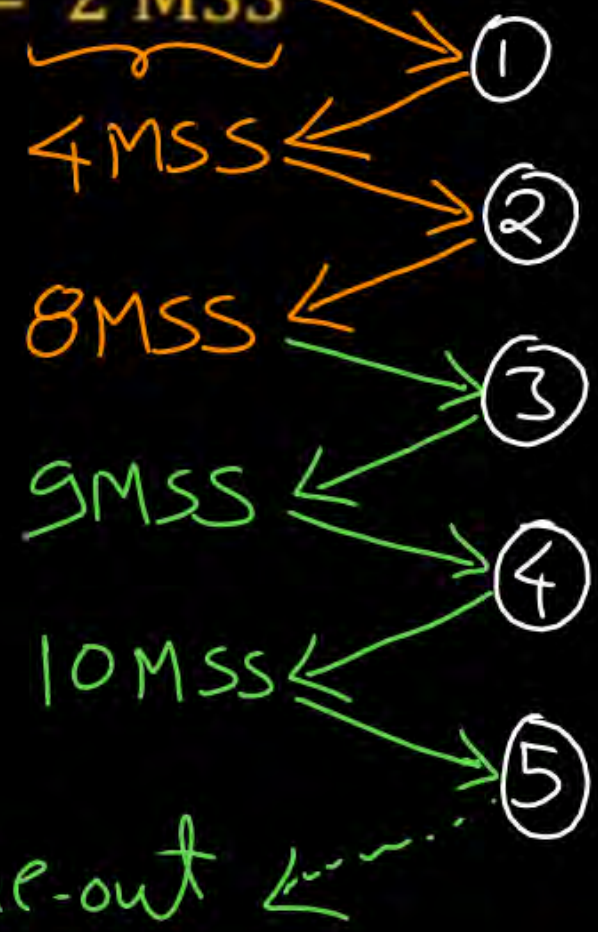
$\rightarrow$ CWND grow linearly // Additive Increase

AIAD
AIMD $\leftarrow$
MIAD
MIMD

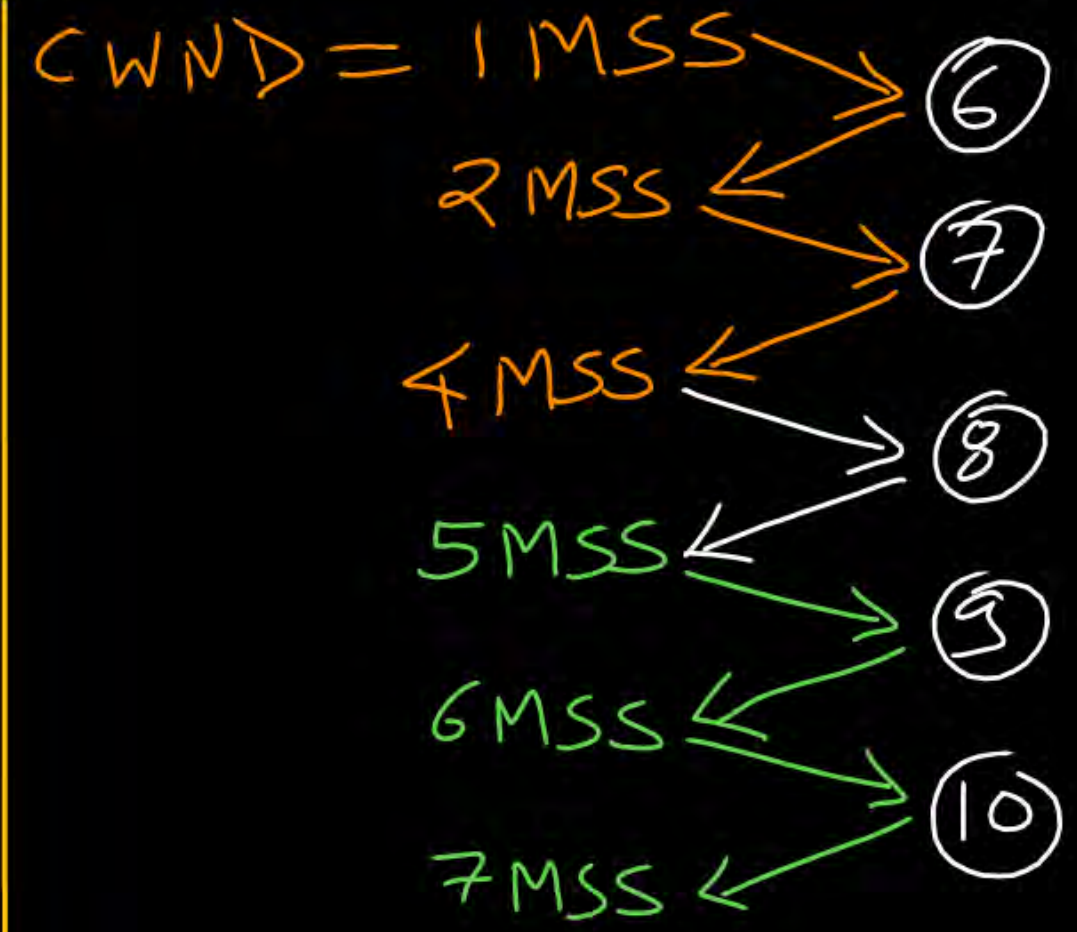
IIT-D $\Rightarrow$
Marks to ALL

SStresh = 8 MSS

CWND = 2 MSS



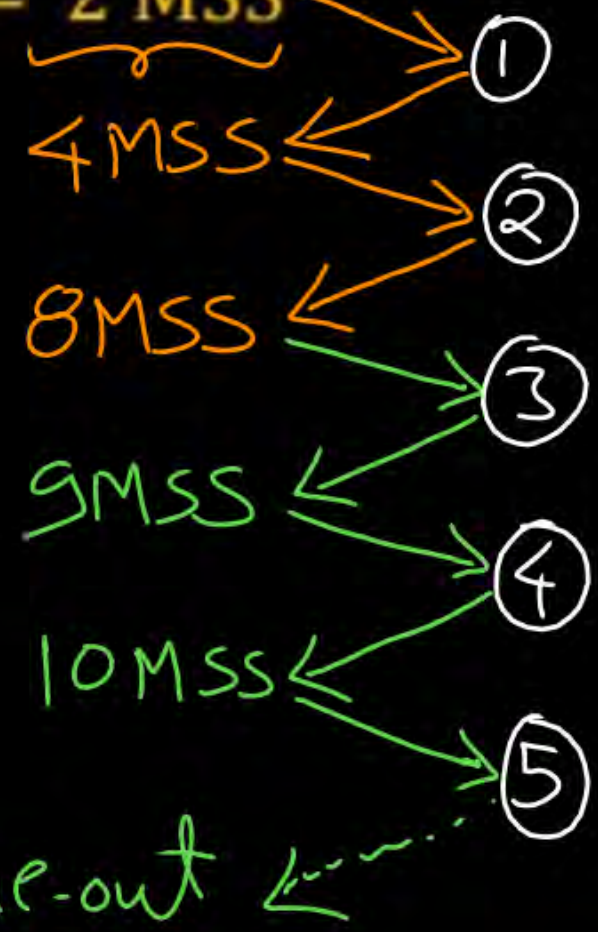
$$\begin{aligned}
 SSTresh &= \frac{CWND}{2} \\
 &= \frac{10 MSS}{2} \\
 &= 5 MSS
 \end{aligned}$$



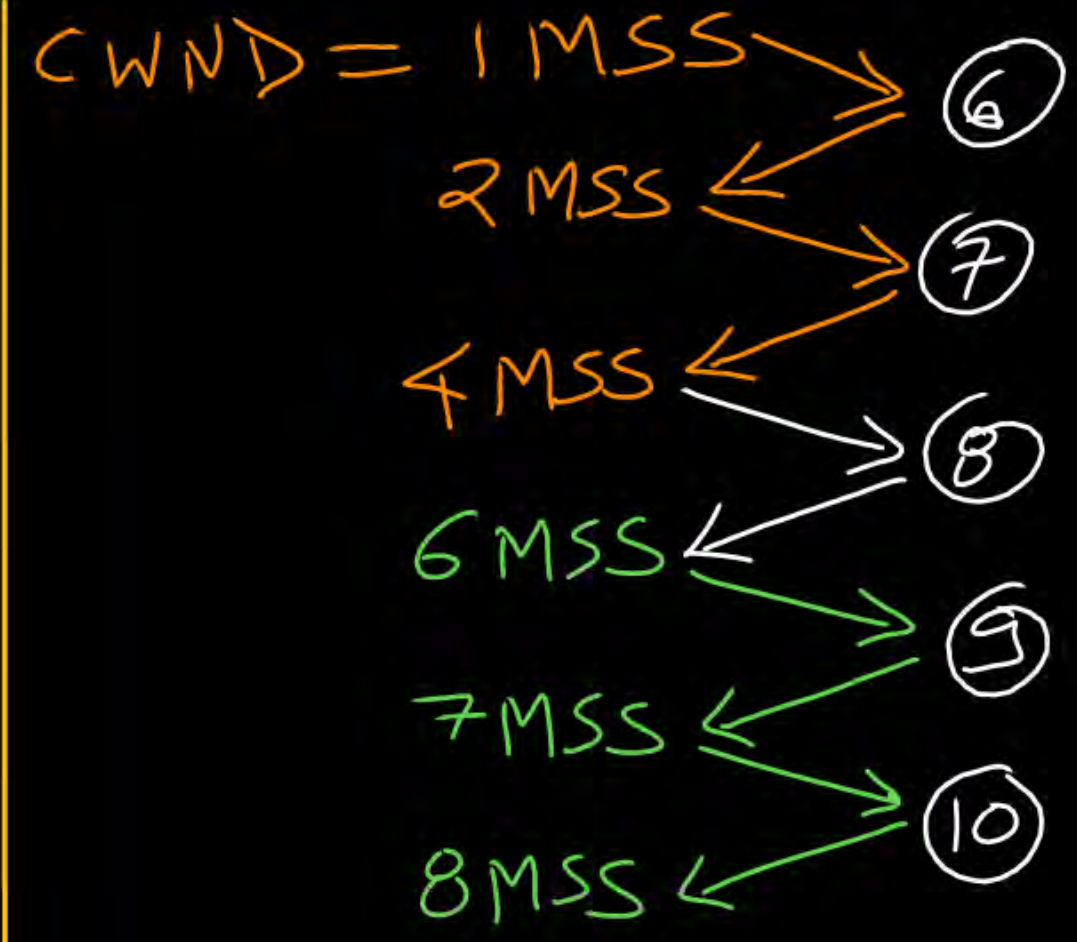
[Ans = 7 MSS]

SStresh = 8 MSS

CWND = 2 MSS



$$\begin{aligned}
 SSTresh &= \frac{CWND}{2} \\
 &= \frac{10 MSS}{2} \\
 &= 5 MSS
 \end{aligned}$$



Ans = 8 MSS

[MCQ]

IIT-D

[GATE-IT-2004] [2 Mark]



#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

- ☐ A 4000 bytes
- ☒ B 8000 bytes
- ☐ C 10000 bytes
- ☐ D 12000 bytes

Ans: B

Send Buffer
(MaxSendBuffer = 12000 Bytes)

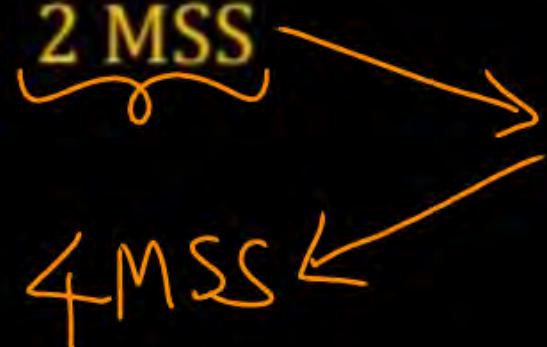


SStresh = Infinite / slow start phase

CWND = 4000 Bytes

MSS = 2000 Bytes

CWND = 4000 Bytes / 2000 Bytes

CWND = 2 MSS

 4 MSS

Ans = 4 MSS
 = 4 * 2000 bytes
 = 8000 byte

Ans = 8000



Topic : Retransmission Policy of TCP

→ Retransmission on time-out

[Retransmit that segment for which time-out occur]

→ Retransmission on receiving '3 duplicate acknowledgments'

[Retransmit the unacked segment with smallest sequence number]

[Do not wait for time-out, "Fast Retransmit", Early retransmit]



Topic : TCP

Client_ISN = 299

MSS = 100 bytes

(P₁) TCP Client

TCP Server

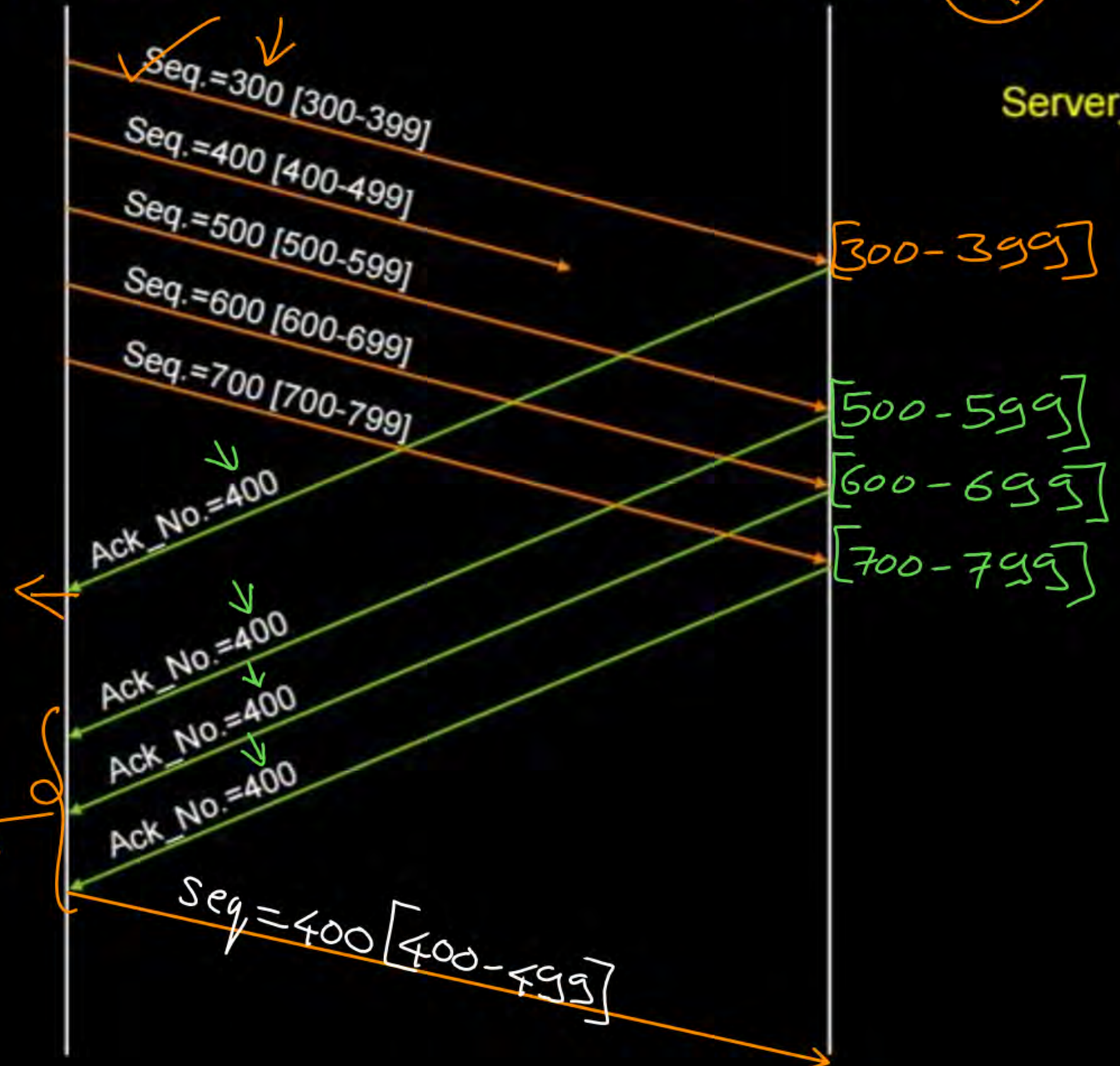
(P₂)



Server_ISN = Y

NBE = ~~300~~
400

Three Duplicate ACK





Topic : Congestion Control



→ In case of "Fast Retransmit" : "Normal Recovery"

Update : SSThresh = $[\text{CWND} / 2]$

CWND = 1 MSS

→ start slow start phase

→ In case of "Fast Retransmit" : "Fast Recovery"

Update : SSThresh = $[\text{CWND} / 2]$

CWND = SSThresh + 3 MSS

→ start congestion Avoidance phase

[MCQ]

11SC

[GATE-IT-2008] [2 Mark]



#Q. Which of the following statements are TRUE?

(S1) TCP handles both congestion and flow control TRUE

(S2) UDP handles congestion but not flow control FALSE

(S3) Fast retransmit deals with congestion but not flow control TRUE

(S4) Slow start mechanism deals with both congestion and flow control FALSE

A S1, S2 and S3 only

B S1 and S3 only

C S3 and S4 only

D S1, S3 and S4 only

S₁ → T/F

S₂ → T/F

Ans: B



Topic : Maximum Segment Lifetime



=> Maximum Segment Lifetime (MSL)

-> Duration for a TCP segment can exist in an inter-network system

=> Wrap-around time

-> Minimum time before the sequence number can be used again

$MSL = \text{Wrap-around time}$

TCP Client Process

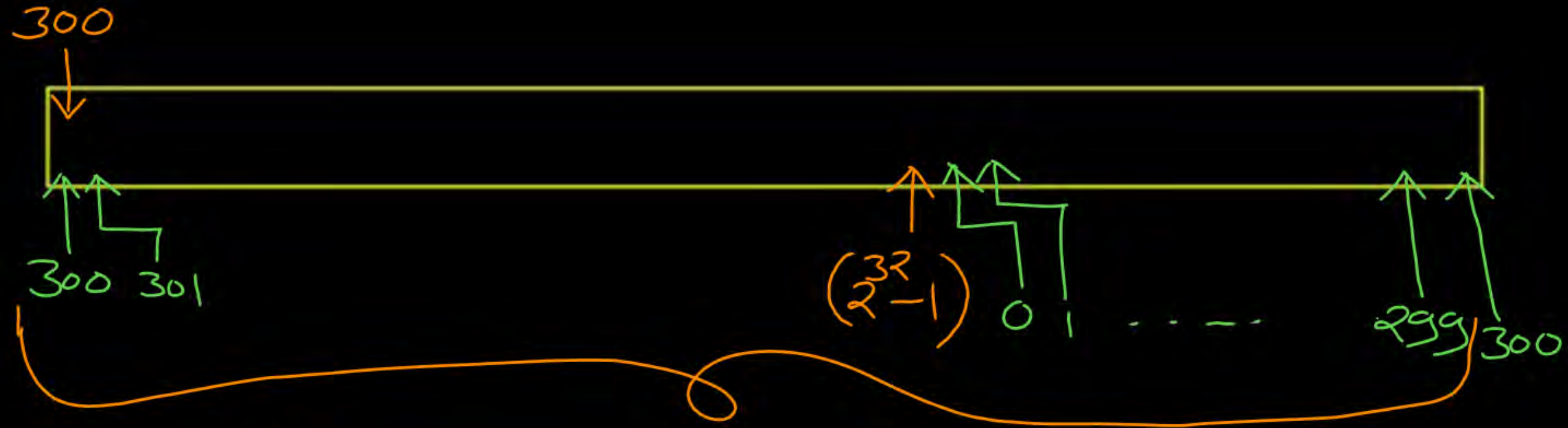
P₁

SOCK_STREAM
(TCP Socket)

Send Buffer

Client_ISN = 299

Seq. No. $\Rightarrow$ 32-bits



2^{32} numbers $\Rightarrow$ 2^{32} bytes (4GB)

$\Rightarrow$ End-to-End Bandwidth



Topic : Maximum Segment Lifetime

$$\Rightarrow K = 32$$



→ Suppose size of sequence number field is k bits in TCP header,
(generally k = 32 bits)

$$\text{MSL} = \frac{2^k \text{ Bytes}}{\text{Bandwidth}} = \frac{2^{32} \text{ Bytes}}{\text{Bandwidth}}$$

$$\text{Wrap-around Time} = \text{MSL}$$

[NAT]

11T-6

[GATE-2018][1 Mark]



#Q. Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps ($= 10^9$ bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

Solution :-



$$\begin{aligned}\text{Wrap-around time (MSL)} &= \frac{2^{32} \text{ Bytes}}{\text{Bandwidth}} = \frac{2^{32} \text{ Bytes}}{16 \text{ bps}} \\ &= \frac{2^{32} \times 2^3 \text{ bits}}{10^9 \text{ bits/sec}} = \frac{2^{35} \text{ bits}}{10^9 \text{ bits/sec}} \\ &= 34.359 \text{ sec} \\ &\approx 34 \text{ sec}\end{aligned}$$

Ans = 34



Topic : Maximum Segment Lifetime



$$\text{Maximum number of bytes can be transmitted in one MSL} = \text{MSL} * \text{Bandwidth}$$

Minimum number of bits required for sequence number field

$$= \lceil \log_2(\text{Max no. of bytes can be transmitted in one MSL}) \rceil \text{ bits}$$

[NAT]

IIT-KGP

[GATE-2022][2 Mark]



#Q. Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.

Ans = 33

Solution :-



$$\begin{aligned}
 \text{Maximum number of bytes can be transmitted in one MSL} &= \text{MSL} * \text{Bandwidth} \\
 &= 60 \text{ SEC} * 1 \text{ Gbps} = 60 \text{ SEC} * 10^9 \text{ bits/sec} \\
 &= 6 * 10^{10} \text{ bits} = \left(\frac{6 * 10^{10}}{8} \right) \text{ bytes}
 \end{aligned}$$

Minimum number of bits required for sequence number field

$$\begin{aligned}
 &= \lceil \log_2(\text{Max no. of bytes can be transmitted in one MSL}) \rceil \text{ bits} \\
 &= \lceil \log_2\left(\frac{6 * 10^{10}}{8}\right) \rceil \text{ bits} = \lceil 32.80 \rceil \text{ bits} = 33 \text{ bits}
 \end{aligned}$$

[MSQ]

IIT-K, H.W.

[GATE-2023][2 Mark]



#Q. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

- ☐ A 30 bits
- ☐ B 32 bits
- ☐ C 34 bits
- ☐ D 36 bits



2 mins Summary



Topic

TCP Congestion Control ✓

Topic

MSL ✓

Topic

~~TCP Connection Close~~



THANK - YOU

